Leakage-Aware Deep Learning for Financial Event-State Forecasting

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ABSTRACT

This project studies aligned company-news event states for multi-horizon excess-return forecasting beyond strong market baselines. I built a local, reproducible artifact pipeline for the top-50 liquid U.S. equities, with same-day-as-close news alignment, no-news states, purged expanding folds, an untouched lockbox, fold-local preprocessing, saved predictions, and tuning ledgers. The primary XGBoost configuration completed 800 development trials and is the strongest lockbox error model: weighted error improved by -5.34×10^{-4} versus the prior XGBoost baseline with a 62.2% both-metric improvement rate. TSMixer is the strongest lockbox portfolio-ranking model, reaching 15.05% over the 137-day lockbox simulation. The Mixed model improved validation by -2.97×10^{-4} versus XGBoost, but worsened lockbox error by $+9.20\times10^{-5}$. The result is objective-specific: XGBoost wins calibrated error forecasting, while TSMixer wins ranking-driven portfolio simulation.

1 INTRODUCTION

Financial news may contain useful short-lived information, but return prediction is low signal-to-noise and highly vulnerable to leakage, multiple testing, and post-hoc narratives. The project therefore frames the task as end-of-day forecasting of forward excess returns for horizons $\{1, 5, 10, 20, 60\}$ days. The contribution is a reproducible deep-learning-ready benchmark: top-50 equities, Alpha Vantage news, Twelve Data OHLCV, explicit no-news rows, purged walk-forward development folds, a final lockbox, and artifact-backed model comparisons. The implemented roster spans linear and tree market baselines, FinBERT feature sidecars, TSMixer sequence models, XGBoost tuning, and horizon-level XGBoost/TSMixer mixing. AUC is intentionally not reported: the target is continuous excess return, so RMSE/MAE, zero-baseline R^2, rank IC, bootstrap support, and lockbox deltas are the relevant metrics.

2 RELATED WORK

The design follows market-efficiency and asset-pricing principles from Fama, Fama-French, and Campbell-Lo-MacKinlay [4, 7, 8]: small apparent edges require strict out-of-sample controls. Textual-finance work shows why news can matter but also why representation matters [15, 17]. Modern NLP motivates contextual encoders and FinBERT-style finance adaptation [2, 6]. For tabular market features, gradient boosting remains a strong baseline [5, 12]. For sequence modeling, the relevant state of the art includes interpretable multi-horizon attention, patch-based transformers, inverted transformers, and foundation time-series models [1, 13, 14, 16]. Evaluation follows forecasting-metric and time-series validation guidance [3, 11]; multiple-testing control follows Harvey et al. [10]. ROC-AUC is cited only to justify exclusion for this regression target [9].

Table 1: Protocol: Leakage-aware artifact pipeline.

Stage	Artifact	Key control	Output
Data	Top-50 + OHLCV + news	no raw overwrite	row universe
Features	market + event states	no-news rows preserved	sidecars
Splits	expanding folds + lockbox	purge/embargo	split manifest
Training	fold-local preprocessing	no lockbox tuning	saved preds
Evaluation	RMSE/MAE + rank IC	bootstrap + lockbox	reports

Horizons: 1d, 5d, 10d, 20d, 60d. Lockbox ends 2026-02-13; all result roots keep manifests and prediction packets.

3 EXPERIMENTS

Data and splits. The benchmark uses 50 liquid equities, five horizons, forward leave-one-out excess-return labels, and explicit event-state rows for no-news dates. The enhanced Alpha Vantage rebuild aligned 166,711 news rows while preserving provider-truncation limitations. The split protocol defines nine expanding walk-forward development folds plus a held-out lockbox from 2025-09-30 to 2026-02-13; lockbox rows are never used for tuning.

Model-selection rationale. The model roster was chosen to separate three questions. First, XGBoost is a strong tabular baseline for sparse market/event features and gives a high bar for calibrated return-level prediction. Second, TSMixer tests whether a 60-day sequence representation can extract temporal structure that a same-row tree model misses. Third, the Mixed model asks whether frozen XGBoost and TSMixer predictions can be combined horizon by horizon without retraining the sequence model. I also built a contextual bandit ranking protocol because the downstream use case is naturally cross-sectional: choose top and bottom baskets, not just minimize pointwise error. That protocol used combined feature and prediction contexts, primary $k = 5$, $\epsilon = 0.05$, 47,250 context rows, and 374,220 selection rows; it guided the ranking-family roadmap while the final comparison here is XGBoost, TSMixer, and the Mixed model.

Models and training. XGBoost is the primary event-market candidate, selected from the full 800-trial Optuna search. The selected trial used 21 features: seven market OHLCV and intraday variables, aligned-news/provider flags, minutes since latest aligned article, eight event-state hash embeddings, and no-news/empty-state flags. Its tree configuration is deliberately small and regularized: 200 histogram trees, depth 2, learning rate 0.0403, subsample 0.987, column subsample 0.543, $\gamma = 0.502$, $\alpha = 7.74$, and $\lambda = 0.151$. The TSMixer incumbent is trial 307 from the sequence branch: 60-day windows, 21 features, two-seed mean, 10 epochs, batch size 2048, four mixer blocks, hidden widths 32/384/32, dropout 0.2, learning rate 0.00401, and 83,017 parameters. The Mixed model is an 800-trial frozen-base horizon mixer over XGBoost and TSMixer predictions; no TSMixer weights were retrained.

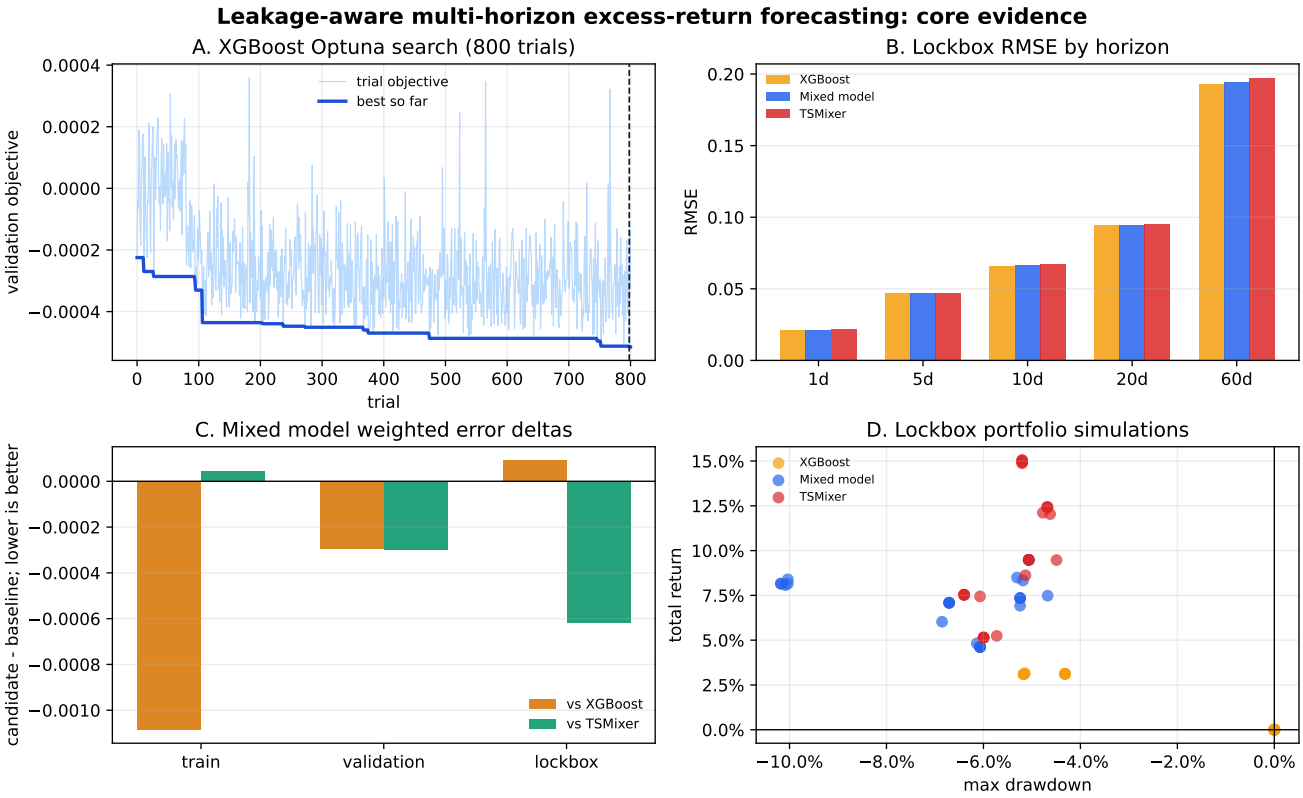


Figure 1: Core evidence. A: XGBoost tuning used train/validation-only selection. B: lockbox RMSE by horizon. C: Mixed model validation gains did not transfer to lockbox. D: the lockbox portfolio simulation summarizes realized forecast rankings over the 137-day evaluation window.

Results. Figure 1 shows the key pattern. XGBoost is the strongest active model on lockbox error metrics: per-horizon RMSE is 0.0214, 0.0468, 0.0661, 0.0943, and 0.1929 for 1d, 5d, 10d, 20d, and 60d respectively (Appendix Table 3). TSMixer is the strongest lockbox ranking model in the portfolio simulation: every best-by-horizon portfolio row is TSMixer, including 15.05% at 1d and 12.43% at 60d (Appendix Table 4). The fuller model and horizon simulation table shows why this is not a contradiction: XGBoost dominates calibrated RMSE and MAE, but its 1d/5d ranking simulations stay at the initial capital with zero trades, whereas TSMixer creates useful ranked baskets across all horizons (Appendix Table 5). The Mixed model improved validation fit but did not beat XGBoost on lockbox error. Thus, error forecasting and cross-sectional ranking produce different winners.

4 DISCUSSION

The strongest lesson is that financial deep learning must be evaluated as a full forecasting system, not as an isolated model leaderboard. XGBoost is the best calibrated error forecaster because it improves the prior XGBoost lockbox weighted error and clears more than half of RMSE/MAE comparisons. Its shallow, strongly regularized trees are well matched to robust point prediction, but that same shrinkage can compress cross-sectional dispersion. TSMixer is the best ranking forecaster because its sequence predictions translate most strongly into lockbox portfolio simulations. The likely mechanism is objective mismatch: RMSE and MAE reward accurate return levels, while the portfolio simulation rewards relative ordering among names on each rebalance date. A sequence model can be worse in absolute error while still placing the better candidates higher in the cross-section. The Mixed model adds scientific value by showing that strong validation performance does not automatically translate to lockbox performance. Next steps are to try a deeper reinforcement learning model with capacity to rank predictions from most probable to least probable, expand data collection beyond the 25 API calls per day bottleneck, and add transaction costs for richer portfolio evaluation. The project therefore produces a strong excess-return forecasting benchmark, an objective-specific model conclusion, and a concrete roadmap for improving the next model family.

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A ARTIFACT ROOTS AND REPLICATION COMMANDS

Repository: https://github.com/ValdiviaEng/financial-event-state-forecasting.git. Primary evidence roots are XGBoost data/v1p0_full_results/v1p0__phase1_top50_updated_av__stage43__xgboost_tuning_20260528T100000Z__src_6dff1bd__run_40f1c9a0/, the Mixed model data/v1p0_full_results/v1p0__phase1_top50_updated_av__stage48__frozen_tsmixer_horizon_mixer__20260529T093301Z__src_6dff1bd__run_fc4c1756/, and the lockbox portfolio simulation data/v1p0_full_results/v1p0__phase1_top50_updated_av__stage47__lockbox_portfolio_simulation__20260529T095157Z__src_6dff1bd__run_8317929b/. Validation command: `source ~/venvs/ai/bin/activate && python -m pytest tests`.

B MODEL AND POLICY CONFIGURATION DETAILS

Table 2: Selected model-family and ranking-policy configuration. The active lockbox comparison uses XGBoost, TSMixer, and the Mixed model; the contextual-bandit protocol records the ranking-oriented direction used for future model selection.

Family	Selection basis	Configuration details
XGBoost	Trial 798 from 800 validation-selected trials	21 event-market features; 200 histogram trees; max_depth 2; learning rate 0.0403; subsample 0.987; colsample_bytree 0.543; colsample_bylevel 0.897; $\gamma = 0.502$; min child weight 9.60; $\alpha = 7.74$; $\lambda = 0.151$; max_bin 512.
TSMixer	Trial 307 sequence incumbent	60-day windows; 21 features; two-seed mean; 10 epochs; batch size 2048; four mixer blocks; hidden time/feature/head widths 32/384/32; dropout 0.2; learning rate 0.00401; weight decay 3.1×10^{-5}; 83,017 parameters.
Mixed model	Trial 391 from 800 validation-selected trials	Frozen-base horizon mixer over XGBoost and TSMixer predictions. Horizon modes: elastic net for 1d, ridge for 5d, convex blends for 10d/20d/60d. TSMixer blend weights were 0.606, 0.323, 0.661, 0.426, and 0.615 for 1d, 5d, 10d, 20d, and 60d.
Contextual bandit	Ranking-policy design	Combined feature/prediction contexts; primary basket size $k = 5$; diagnostic $k \in \{3, 10\}$; primary $\epsilon = 0.05$; diagnostic $\epsilon \in \{0.01, 0.10\}$; 47,250 context rows; 374,220 selection rows; 11 policy variants.

C EXACT LOCKBOX RESULTS

Table 3: Lockbox metrics by model and horizon. Lower RMSE/MAE is better; R^2 is against the zero-return baseline.

Model	Horizon	RMSE	MAE	R^2
XGBoost	1d	0.0214	0.0146	0.0000
TSMixer	1d	0.0221	0.0154	-0.0640
Mixed model	1d	0.0214	0.0146	-0.0000
XGBoost	5d	0.0468	0.0335	0.0000
TSMixer	5d	0.0469	0.0337	-0.0085
Mixed model	5d	0.0469	0.0336	-0.0058
XGBoost	10d	0.0661	0.0488	0.0000
TSMixer	10d	0.0670	0.0494	-0.0262
Mixed model	10d	0.0665	0.0490	-0.0101
XGBoost	20d	0.0943	0.0696	-0.0014
TSMixer	20d	0.0949	0.0701	-0.0148
Mixed model	20d	0.0944	0.0696	-0.0031
XGBoost	60d	0.1929	0.1396	0.0072
TSMixer	60d	0.1972	0.1398	-0.0374
Mixed model	60d	0.1942	0.1384	-0.0063

Table 4: Best portfolio row by horizon over the 137-day lockbox interval. The rows summarize realized forecast-ranking simulations under the stated frictionless assumptions.

Horizon	Model / policy	Return	Final equity	Trades
1d	TSMixer / one_d_bot10	15.05%	$115,045	1161
5d	TSMixer / one_d_bot10	5.24%	$105,240	1831
10d	TSMixer / none	9.48%	$109,482	1780
20d	TSMixer / none	7.54%	$107,536	1680
60d	TSMixer / botk_3_of_5	12.43%	$112,434	1319

Table 5: Best lockbox simulation row for each model and horizon. Initial capital is $100,000. Turnover is total traded notional divided by initial capital.

Model	Horizon	Best policy	Return	Final equity	Max DD	Trades	Turnover
XGBoost	1d	one_d_bot10	0.00%	$100,000	0.00%	0	0.0x
XGBoost	5d	one_d_bot10	0.00%	$100,000	0.00%	0	0.0x
XGBoost	10d	one_d_bot10	0.00%	$100,000	0.00%	0	0.0x
XGBoost	20d	botk_2_of_5	3.16%	$103,160	-5.14%	1681	4.7x
XGBoost	60d	one_d_bot10	3.13%	$103,125	-4.32%	1280	1.4x
TSMixer	1d	one_d_bot10	15.05%	$115,045	-5.20%	1161	34.0x
TSMixer	5d	one_d_bot10	5.24%	$105,240	-5.73%	1831	21.8x
TSMixer	10d	none	9.48%	$109,482	-5.07%	1780	10.0x
TSMixer	20d	none	7.54%	$107,536	-6.40%	1680	4.7x
TSMixer	60d	botk_3_of_5	12.43%	$112,434	-4.69%	1319	1.4x
Mixed model	1d	botk_4_of_5	8.39%	$108,393	-10.03%	1267	54.1x
Mixed model	5d	one_d_bot10	4.82%	$104,820	-6.14%	1830	21.9x
Mixed model	10d	none	9.48%	$109,482	-5.07%	1780	10.0x
Mixed model	20d	botk_2_of_5	7.09%	$107,090	-6.71%	1680	4.8x
Mixed model	60d	one_d_bot10	7.48%	$107,483	-4.68%	1456	1.6x

D SUPPLEMENTARY VISUALIZATIONS

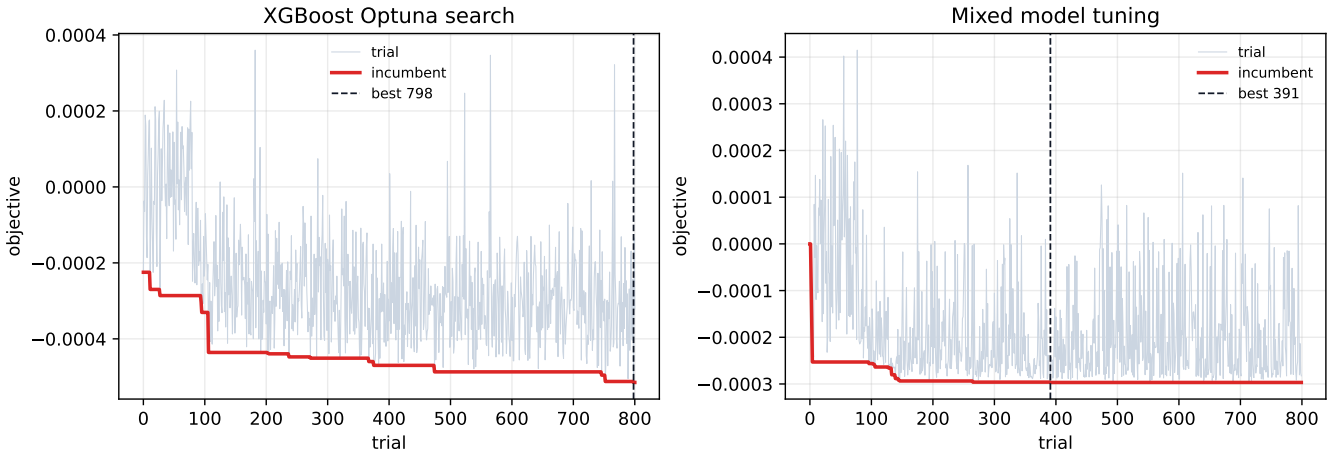


Figure 2: Optuna incumbent curves for XGBoost and the Mixed model.

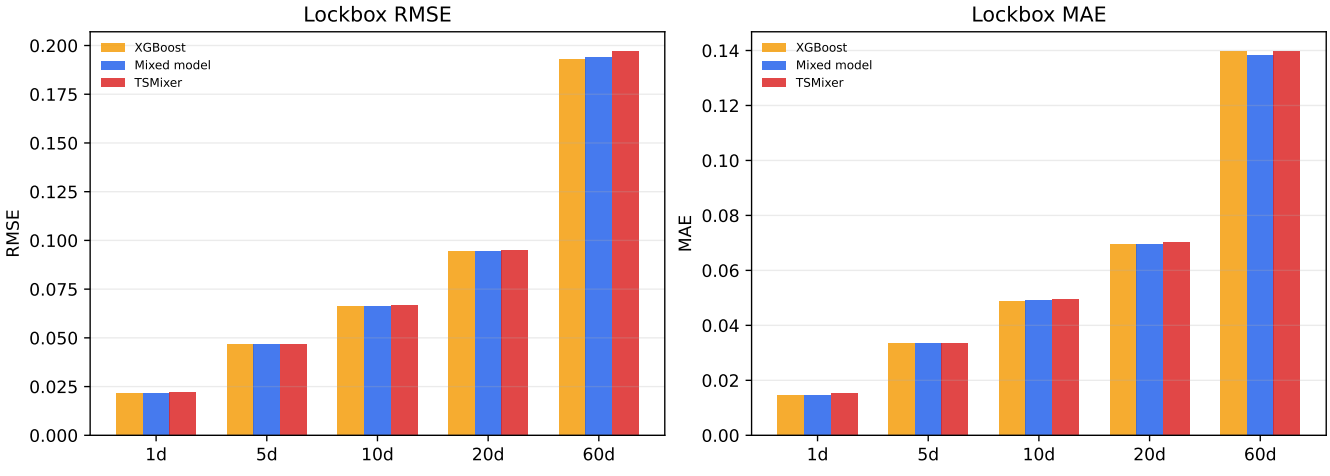


Figure 3: Lockbox RMSE and MAE by horizon for XGBoost, TSMixer, and the Mixed model.

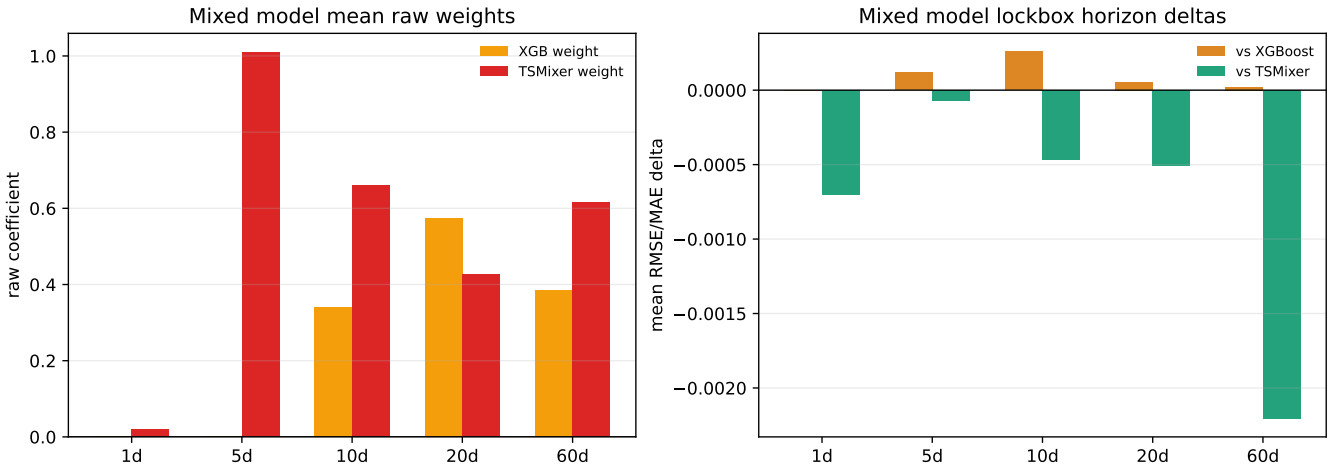


Figure 4: Horizon-specific raw mixer weights and lockbox horizon deltas.

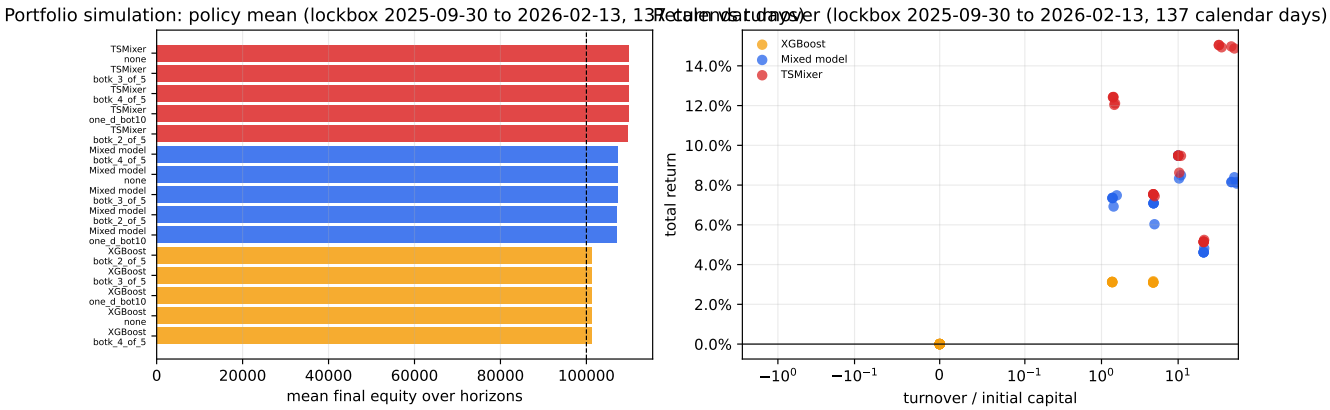


Figure 5: Portfolio simulations over the 2025-09-30 to 2026-02-13 lockbox interval (137 calendar days): final equity by policy and return versus turnover.

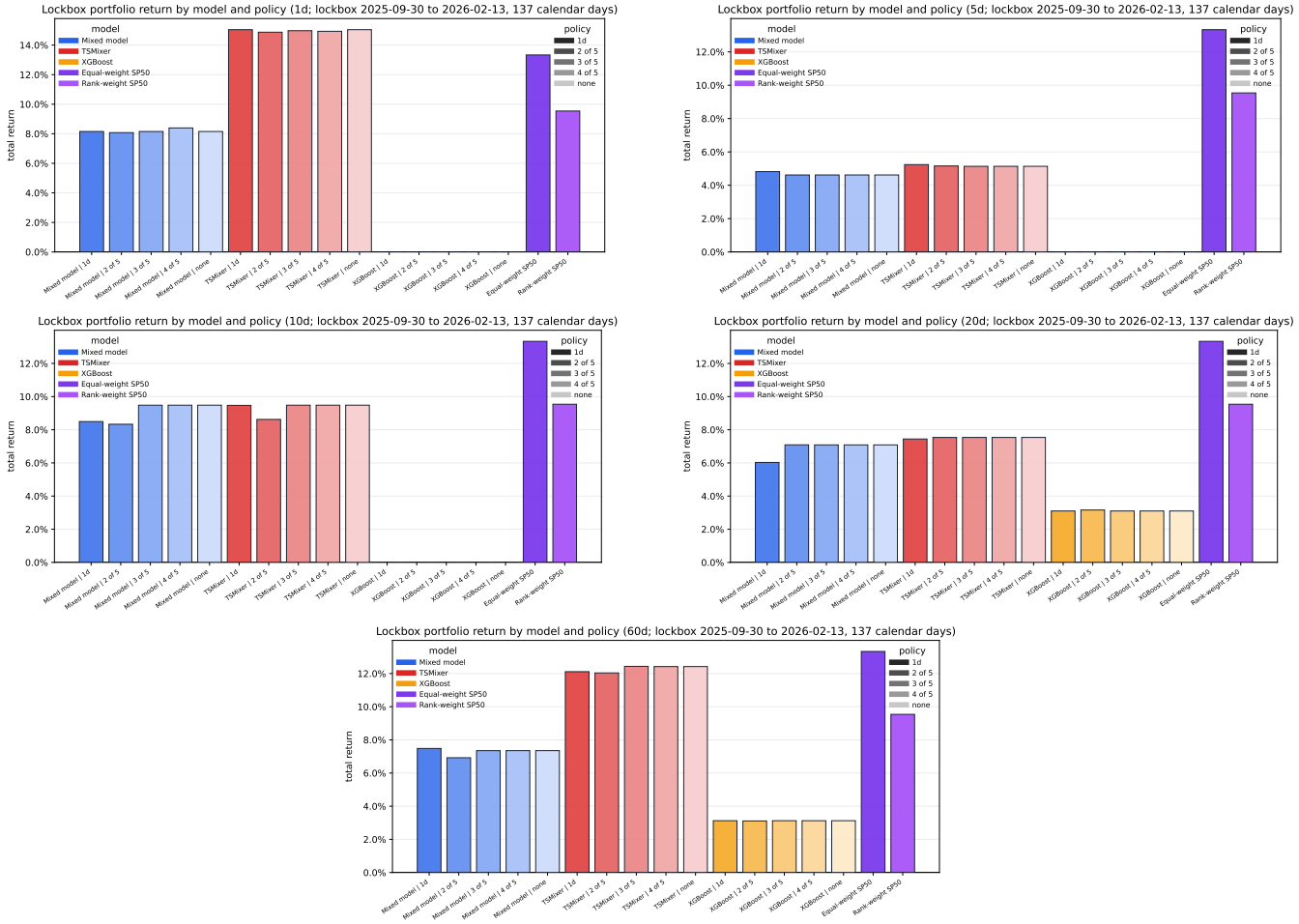


Figure 6: Portfolio return bars by horizon, model, sell policy, and SP50 benchmark proxy over the 137-day lockbox interval.

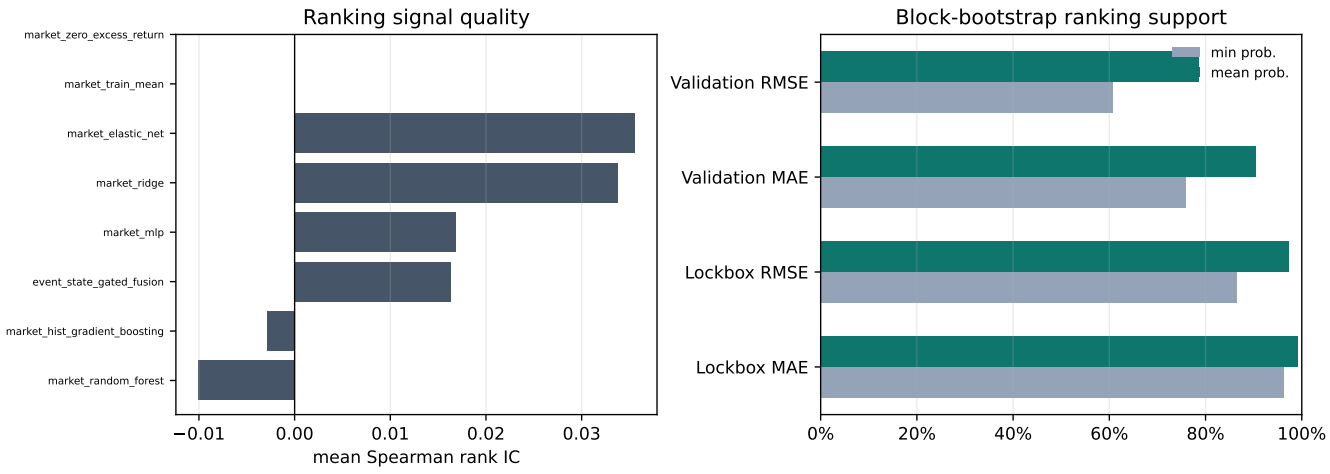


Figure 7: Ranking signal quality and block-bootstrap support.